

Appendix: TCT Model Sources and Results

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A Purpose

This appendix records the material needed to reproduce the numerical results quoted in the report: the five model sources as they were filed into TCT, the TCT commands in the order they were issued, the sizes and control data TCT returned, and a checklist against which a re-run can be verified.

B TCT operation steps

Start TCTX64_20210701.exe in the folder that contains the USER directory. Each model is a plain-text .ADS file; FD converts it to a binary .DES model. The commands below are the menu entries used, in order.

```
1. FD          file all .ADS to .DES      -> UAV1C, UAV2C, TEAM, SPCOVRC, SPFIFO
2. 3 Sync      k = 3 : UAV1C, UAV2C, TEAM -> PLANTC   (45,140)
3. 4 Meet      k = 2 : SPCOVRC, SPFIFO    -> SPECC    (20,234)
4. 5 Supcon     PLANTC, SPECC              -> SUPC     (46,128)
5. 7 Condat     PLANTC, SUPC               -> DATC     Controllable.
6. FE PLANTC / FE SUPC / FA DATC          export the PDS and PDT text files
```

C Model sources

The five .ADS files are reproduced verbatim below. In each file, odd event labels are controllable and even labels are uncontrollable, as TCT requires.

UAV1C.ADS - UAV1 plant model.

```
# Design C UAV1 plant model.
# State 3 = down (fault 4 or loss of contact 24), awaiting ground recovery.
# Odd labels are controllable, even labels uncontrollable.
UAV1C
State size
6
Marker states:
0
1
Vocal states:

Transitions:
0 1 1
1 3 0
1 2 2
1 4 3
1 24 3
3 5 3
3 6 0
2 8 4
4 9 5
4 11 1
5 10 1
```

UAV2C.ADS - UAV2 plant model.

```
# Design C UAV2 plant model.
# State 3 = down (fault 16 or loss of contact 26), awaiting ground recovery.
# Odd labels are controllable, even labels uncontrollable.
UAV2C
State size
6
Marker states:
0
1
Vocal states:

Transitions:
0 13 1
1 15 0
1 14 2
1 16 3
1 26 3
3 17 3
3 18 0
2 20 4
4 21 5
4 23 1
5 22 1
```

TEAM.ADS - Ground rescue team.

```

# Design C ground rescue team: ONE team with two duties.
# State 0 idle, 1 rescuing, 2 repairing UAV1, 3 repairing UAV2.
# Transitions 2-21-1 and 3-9-1 abort a repair in order to answer a call.
TEAM
State size
4
Marker states:
0
Vocal states:

Transitions:
0 9 1
0 21 1
2 21 1
3 9 1
1 10 0
1 22 0
0 5 2
2 6 0
0 17 3
3 18 0

```

SPCOVRC.ADS - Specification 1: coverage preservation.

```

# Design C coverage requirement, fully self-looped.
# Keep at least one UAV exploring whenever a controllable choice allows it.
# Loss of contact ends exploration like a fault. All states marked.
SPCOVRC
State size
4
Marker states:
0
1
2
3
Vocal states:

```

```

Transitions:
0 1 1
2 1 3
3 3 2
1 2 0
3 2 2
1 4 0
3 4 2
0 10 1
2 10 3
0 11 1
2 11 3
0 13 2
1 13 3
3 15 1
2 14 0
3 14 1
2 16 0
3 16 1
0 22 2
1 22 3
0 23 2
1 23 3
1 24 0
3 24 2
2 26 0
3 26 1
0 5 0
0 6 0
0 8 0
0 9 0
0 17 0
0 18 0
0 20 0
0 21 0
1 5 1
1 6 1
1 8 1
1 9 1
1 17 1
1 18 1
1 20 1
1 21 1
2 5 2
2 6 2
2 8 2
2 9 2

```

```

2 17 2
2 18 2
2 20 2
2 21 2
3 5 3
3 6 3
3 8 3
3 9 3
3 17 3
3 18 3
3 20 3
3 21 3

```

SPFIFO.ADS - Specification 2: first faulted first repaired.

```

# Design C requirement, fully self-looped.
# If both UAVs are down, the one that faulted first must be repaired first.
# 0 neither down, 1 only UAV1, 2 only UAV2,
# 3 both (UAV1 faulted first), 4 both (UAV2 faulted first).
# Event 17 is absent in state 3 and event 5 in state 4. All states marked.
SPFIFO

```

State size

5

Marker states:

0

1

2

3

4

Vocal states:

Transitions:

```

0 4 1
0 24 1
0 16 2
0 26 2
1 16 3
1 26 3
1 6 0
2 4 4
2 24 4
2 18 0
3 6 2
4 18 1
0 5 0
0 17 0
1 5 1
1 17 1
2 5 2
2 17 2
3 5 3
4 17 4
0 1 0
0 2 0
0 3 0
0 8 0
0 9 0
0 10 0
0 11 0
0 13 0
0 14 0
0 15 0
0 20 0
0 21 0
0 22 0
0 23 0
1 1 1
1 2 1
1 3 1
1 8 1
1 9 1
1 10 1
1 11 1
1 13 1
1 14 1
1 15 1
1 20 1
1 21 1
1 22 1
1 23 1
2 1 2
2 2 2
2 3 2
2 8 2

```

```

2 9 2
2 10 2
2 11 2
2 13 2
2 14 2
2 15 2
2 20 2
2 21 2
2 22 2
2 23 2
3 1 3
3 2 3
3 3 3
3 8 3
3 9 3
3 10 3
3 11 3
3 13 3
3 14 3
3 15 3
3 20 3
3 21 3
3 22 3
3 23 3
4 1 4
4 2 4
4 3 4
4 8 4
4 9 4
4 10 4
4 11 4
4 13 4
4 14 4
4 15 4
4 20 4
4 21 4
4 22 4
4 23 4

```

D Results reported by TCT

Sizes as printed by TCT, followed by the control data from the Condat output, which lists the supervisor states at which a controllable event must be disabled.

```

UAV1C = Create(...)           (6, 11)
UAV2C = Create(...)           (6, 11)
TEAM  = Create(...)           (4, 10)
PLANTC = Sync(UAV1C,UAV2C,TEAM) (45, 140) Blocked_events = None
SPECC  = Meet(SPCOVRC,SPFIFO)  (20, 234)
SUPC   = Supcon(PLANTC,SPECC)  (46, 128)
DATC   = Condat(PLANTC,SUPC)    Controllable.

```

Condat control data (DATC PDT):

Control data are displayed as a list of supervisor states where disabling occurs, together with the events that must be disabled there.

```

control data:
1: 3           2: 15
9: 15          11: 15
12: 3          13: 3
17: 15         20: 15
22: 17         23: 3
24: 5          25: 3
27: 15         35: 3

```

E Reproduction checklist

Sync(UAV1C,UAV2C,TEAM) must report (45,140) and Blocked_events = None.

Meet(SPCOVRC,SPFIFO) must report (20,234).

Supcon(PLANTC,SPECC) must report (46,128).

Condat(PLANTC,SUPC) must report Controllable.

The control data must list 14 state-event pairs: event 3 six times, event 15 six times, event 5 once and event 17 once.

If any of these differs, the model has been changed since this submission. The complete TCT session log, including every command and every reply, is provided alongside this appendix as TCT_SESSION_MACRO.txt.